IEOR 4574E001 · AI ENGINEERING APPS

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LESSON 01

Introduction to AI Engineering

01

Ciro Greco

Columbia University · 2026

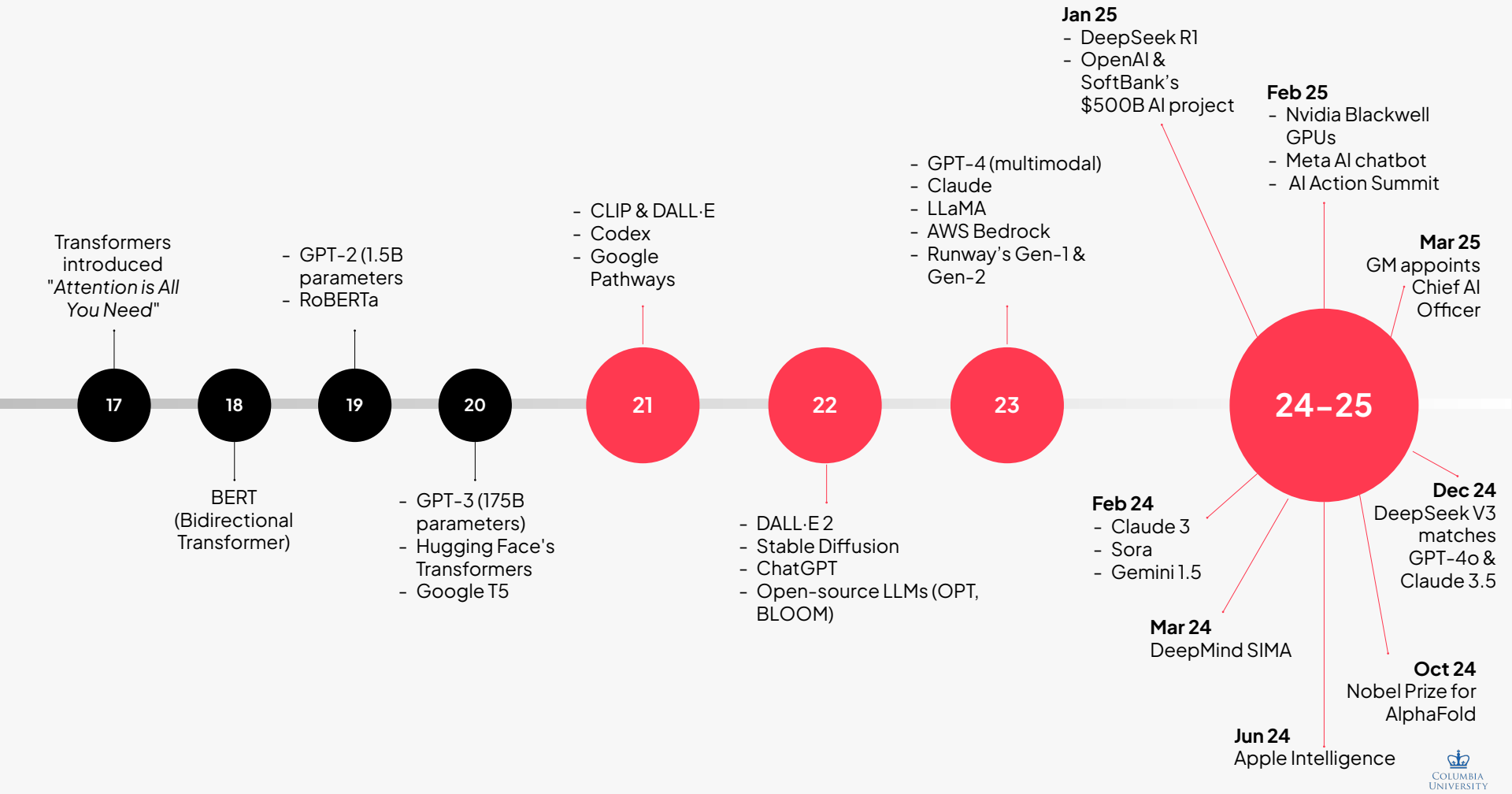


Nice to meet you

- Co-founder and CEO at **Bauplan**.
- Previously co-founder at Tooso (acquired by Coveo TSX:CVO).
- VP AI at Coveo from scale-up to IPO.
- Postdoc in computational stuff.

Large Language Models

- LLMs = very large **deep learning** models that are **pre-trained** on vast amounts of data to predict the next **token** in a sequence.
- Based on transformer architecture (*Vaswani et al., 2017*).
- Capabilities: reasoning, summarization, translation, code generation, tool use.
- Examples: GPT-6.1 Sol, Fable 5.1, Gemini 4 Carbon.


Transformers introduced "*Attention is All You Need*"
17
18
BERT (Bidirectional Transformer)
19
- GPT-2 (1.5B parameters
- RoBERTa
20
- GPT-3 (175B parameters)
- Hugging Face's Transformers
- Google T5
21
- CLIP & DALL·E
- Codex
- Google Pathways
22
- DALL·E 2
- Stable Diffusion
- ChatGPT
- Open-source LLMs (OPT, BLOOM)
23
- GPT-4 (multimodal)
- Claude
- LLaMA
- AWS Bedrock
- Runway's Gen-1 & Gen-2
24–25
Jan 25
- DeepSeek R1
- OpenAI & SoftBank's $500B AI project
Feb 25
- Nvidia Blackwell GPUs
- Meta AI chatbot
- AI Action Summit
Mar 25
GM appoints Chief AI Officer
Dec 24
DeepSeek V3 matches GPT-4o & Claude 3.5
Oct 24
Nobel Prize for AlphaFold
Jun 24
Apple Intelligence
Mar 24
DeepMind SIMA
Feb 24
- Claude 3
- Sora
- Gemini 1.5
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AI in 2026

It’s a fantastic moment to be in AI

53%

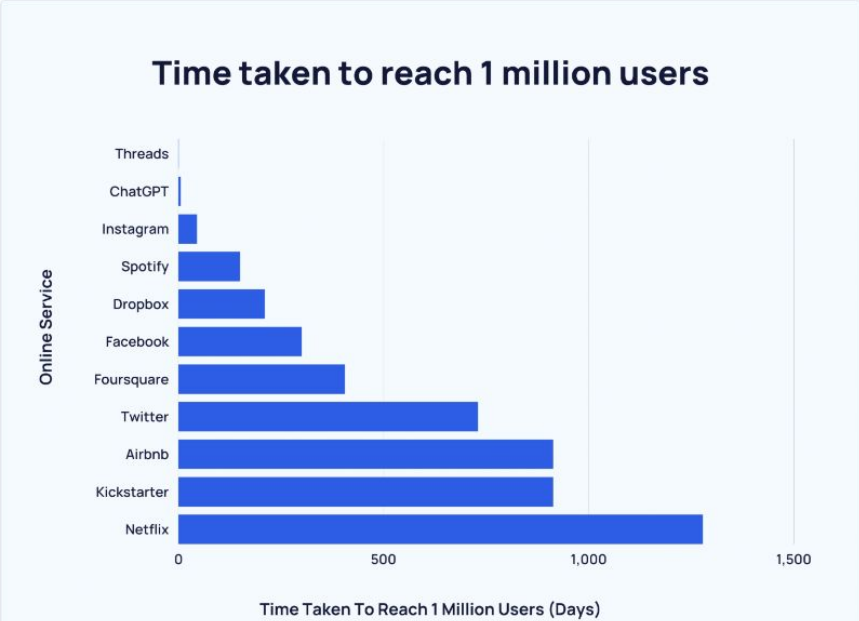
Population Adoption

Reached within just three years, spreading significantly faster than either the PC or the internet.*

$172B

Annual Consumer Value

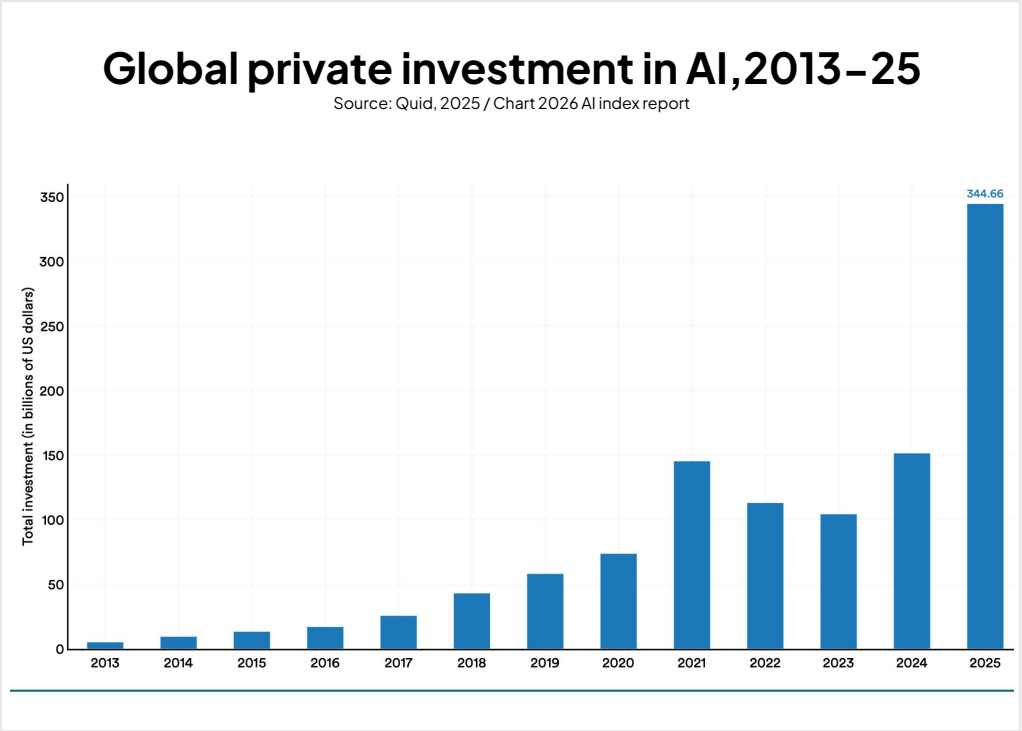
The estimated annual value of generative AI tools to U.S. consumers by early 2026, with the median value per user tripling between 2025 and 2026.*



*Artificial Intelligence Index Report 2026 | Chart from Huyen 2025

Global private investment in AI, 2025

- **$345B** in private investments, up 127.5% on 2024.
- Model layer's commoditization:
  - Models become cheaper over time.
  - Capabilities and benchmarks tend to converge.
  - Switching and interoperating between models becomes easier .



AI enterprise adoption in 2025

$37B total, 2025 in the US

Applications $19B
Infrastructure $18B

Copilots $7.2B
Coding $4.0B
Healthcare $1.5B
Foundation-model APIs $12.5B

WHERE IT IS USED

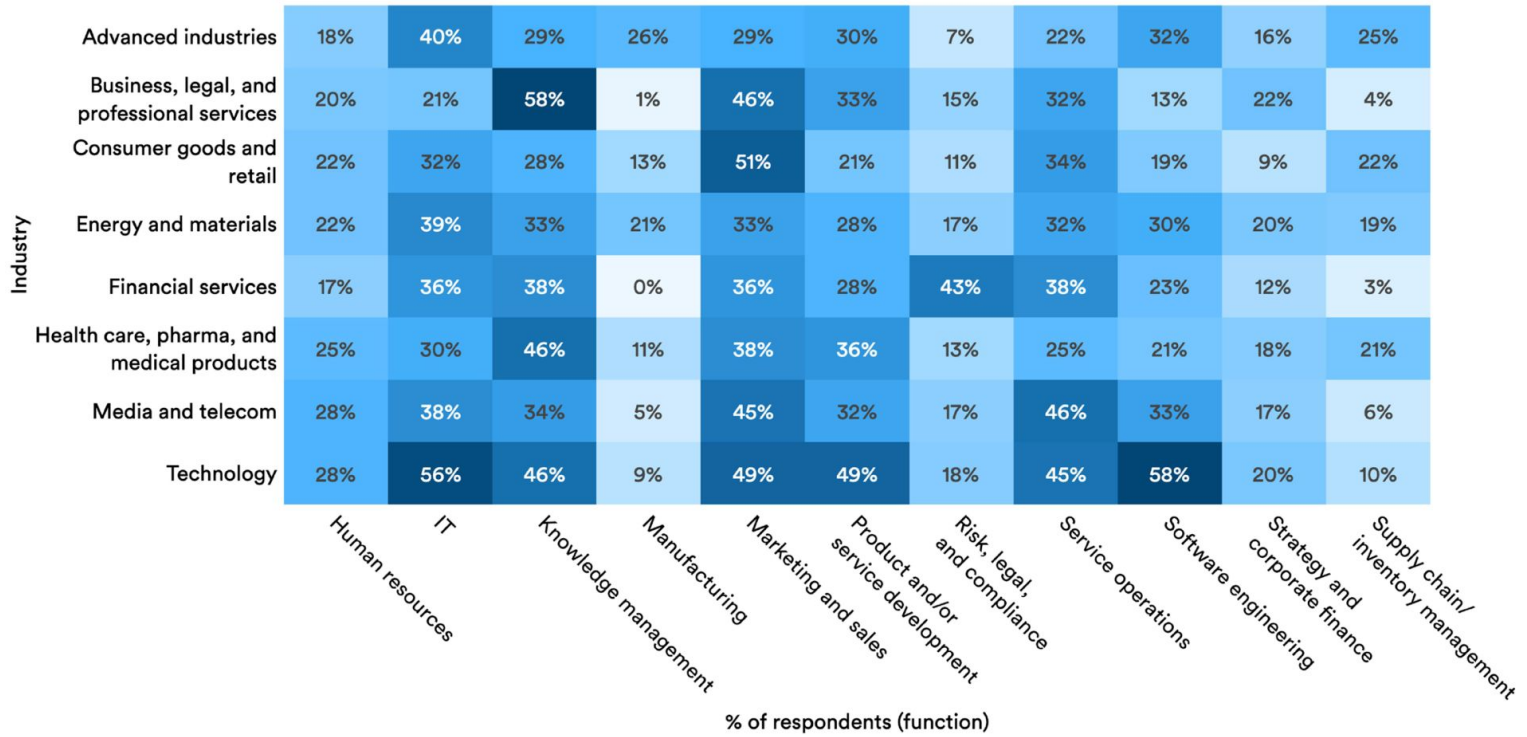
% of firms using AI in that function

Software engineering	**58%**
Knowledge management	**58%**
IT	**56%**
Marketing & sales	**51%**
Product development	**49%**
Service operations	**46%**
Risk, legal & compliance	**43%**

Spend: Menlo Ventures 2025 (U.S. enterprise, $37B total). Usage: McKinsey via AI Index 2026 Fig. 4.3.3 — technology-sector row. Agent depth: Figs. 4.3.7–4.3.8.

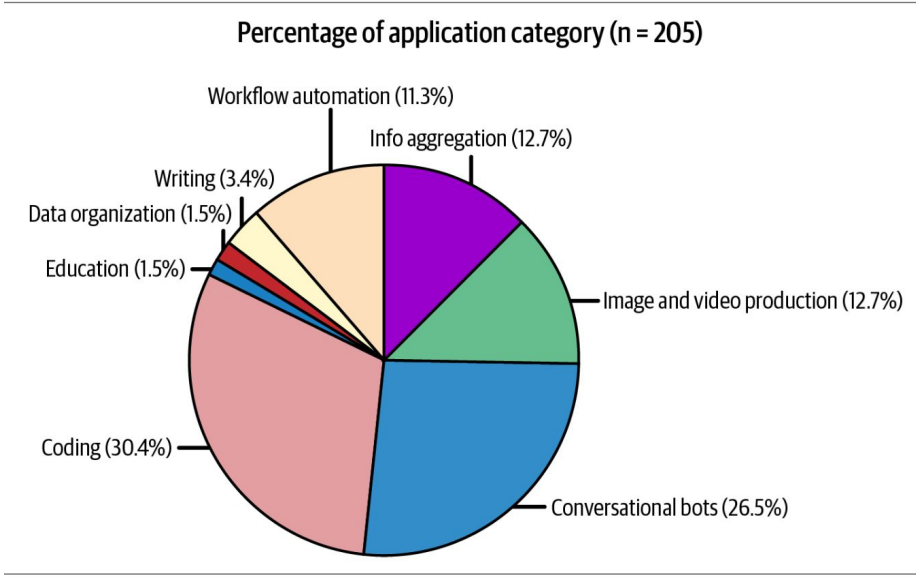
AI use by industry and function 2025

Source: McKinsey & Company / Chart 2026 AI index report



Applications

- Code generation & IDE copilots (B2B)
- General-purpose chat assistants (B2C)
- Image generation & editing
- Customer support AI agents
- Enterprise productivity
- Video generation
- Design/marketing content creation



re 1-7. Distribution of use cases in the 205 open source repositories on GitHub

Image from Huyen 2025

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AI's impact and the labor market

Negative impact

−20%

employed software developers aged 22–25, vs their 2022 peak.
Senior software engineers grew.

And ahead: 1 in 3 employers expect AI to shrink their headcount within a year. 43% expect no change.

Positive impact

~1,000,000

U.S. job postings in 2025 that required AI skills (about 1 in every 40 postings in the country).

Information 13.2% of postings (was 7.8% last year) · Finance 5.3% · Manufacturing 4.7% — and rising in transport, real estate and education.

Asking for: cloud · scalability · orchestration · agents **Not any more:** "ChatGPT" · "chatbot" · "conversational AI"

Sources: AI Index 2026, ch. 4: Figs. 4.4.1, 4.4.9, 4.4.10–11 (postings) · 4.4.29–4.4.31 (employment, Brynjolfsson et al. 2025) · 4.4.33–34 (McKinsey expectations).

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The rise of AI engineering

Shift towards production

Moving from experimental AI research to execution. The primary goal is now building reliable, production-ready, and scalable applications.

Core Focus Areas

1. Specification and evaluation.
2. Model selection and cost engineering.
3. Context engineering.
4. Agents and tool use.
5. Deployment and observability.
6. Product development.

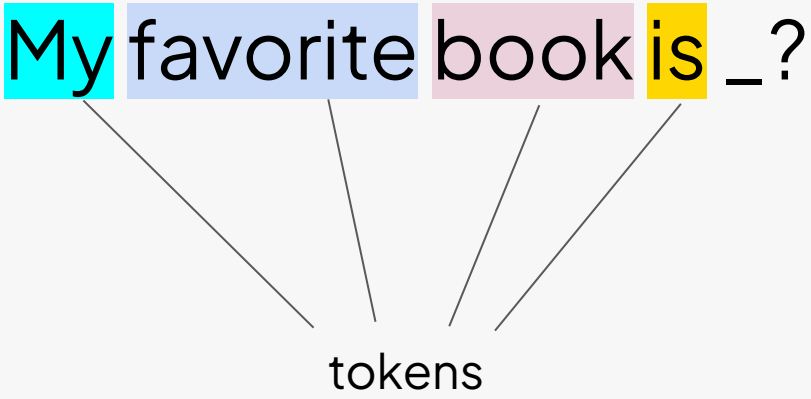
AI engineering is to AI research what Software Engineering is to Computer Science.

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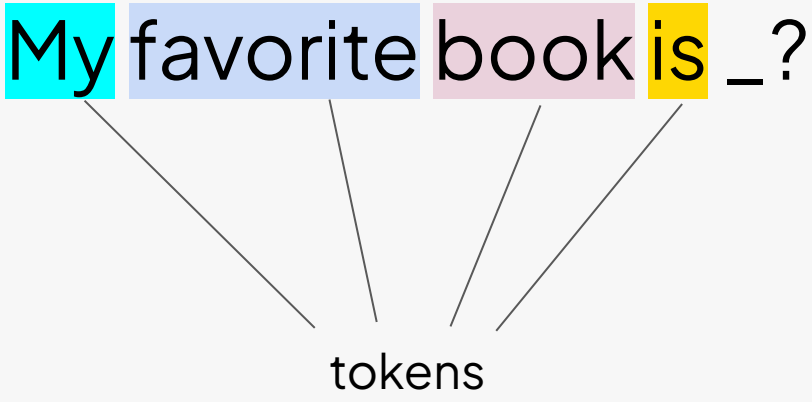
Large Language Models

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Language models are completion machines



Language models are completion machines



Moby Dick	4.1%
War and Peace	3.9%
the one	1.4%
written	2.2%

Pick on of these, but with a bit of randomness

argmax ---------- **sampling**

temperature

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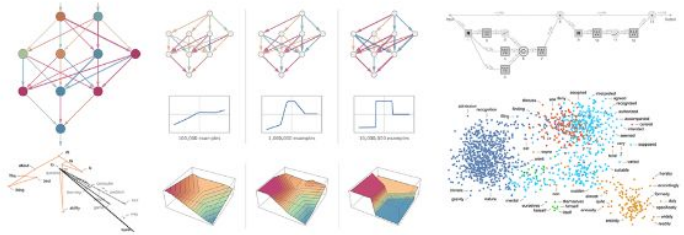
Language models are completion machines

- Both traditional LMs and today's LLMs do the same core thing: **predict the next token given context**.
- More specifically: they decode by **sampling** or by **taking argmax** to complete a sequence.

Stephen Wolfram introduction to LLM

What Is ChatGPT Doing ... and Why Does It Work?

February 14, 2023



It's Just Adding One Word at a Time

That ChatGPT can automatically generate something that reads even superficially like human-written text is remarkable, and unexpected. But how does it do it? And why does it work? My purpose here is to give a rough outline of what's going on inside ChatGPT—and then to explore why it is that it can do so well in producing what we might consider to be meaningful text. I should say at the outset that I'm going to focus on the big picture of what's going on—and while I'll mention some engineering details, I won't get deeply into them. (And the essence of what I'll say applies just as well to other current "large language models" [LLMs] as to ChatGPT.)

The first thing to explain is that what ChatGPT is always fundamentally trying to do is to produce a "reasonable continuation" of whatever text it's got so far, where by "reasonable" we mean "what one might expect someone to write after seeing what people have written on billions

So what's different with LLMs?

Scale
Much larger parameters, data, and compute → richer learned distributions, emergent capabilities.

Generality
Compared to n-grams and RNN/LSTM, transformer based LLMs can perform a variety of tasks by simply changing the context.

Training recipe
Self-supervised pretraining + instruction tuning + preference optimization (RLHF/DPO) → chat/task behavior.

Training recipe: self-supervision

- **Self-supervised learning:** training on massive unlabeled corpora.
- **Cost is exponentially lower**: no human labels → unprecedented scaling becomes possible.
- Data = internet text, code, images.
- Changed how we think about datasets: **quantity and diversity > handcrafted labels.**

Input: "The cat sat on the **[MASK]"**

Label: "mat" (comes directly from the original sentence).

<BOS>	the
<BOS>, the,	cat
<BOS>, the, cat	sat
<BOS>, the, cat, sat	on
<BOS>, the, cat, sat, on	the
<BOS>, the, cat, sat, on, the	mat

Training set

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Scale

Parameters

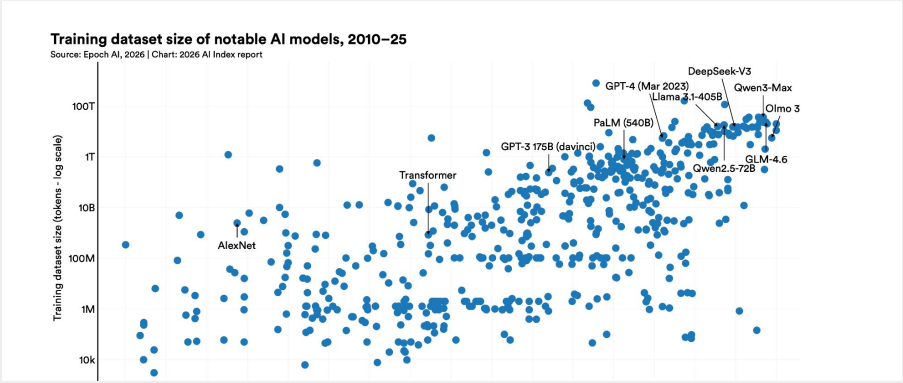
Flat for three years — near 1 trillion but OpenAI, Anthropic and Google no longer disclose parameter counts, dataset sizes or training duration at all.

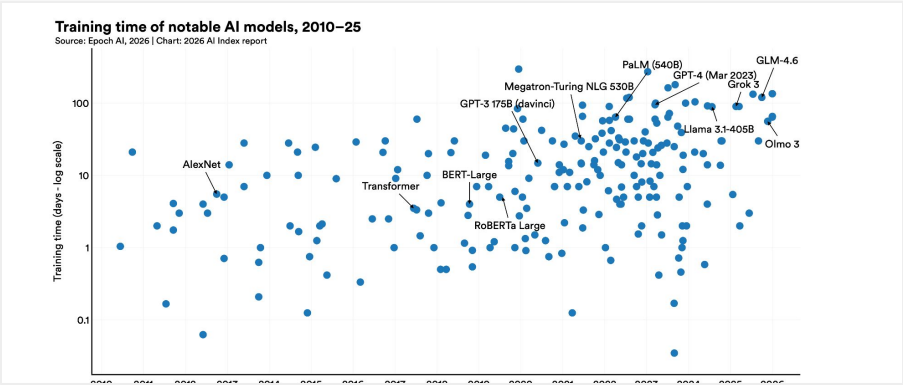
Compute

Still climbing, by orders of magnitude. Global AI compute grew 3.3× per year since 2022, reaching 17.1 million H100-equivalents.

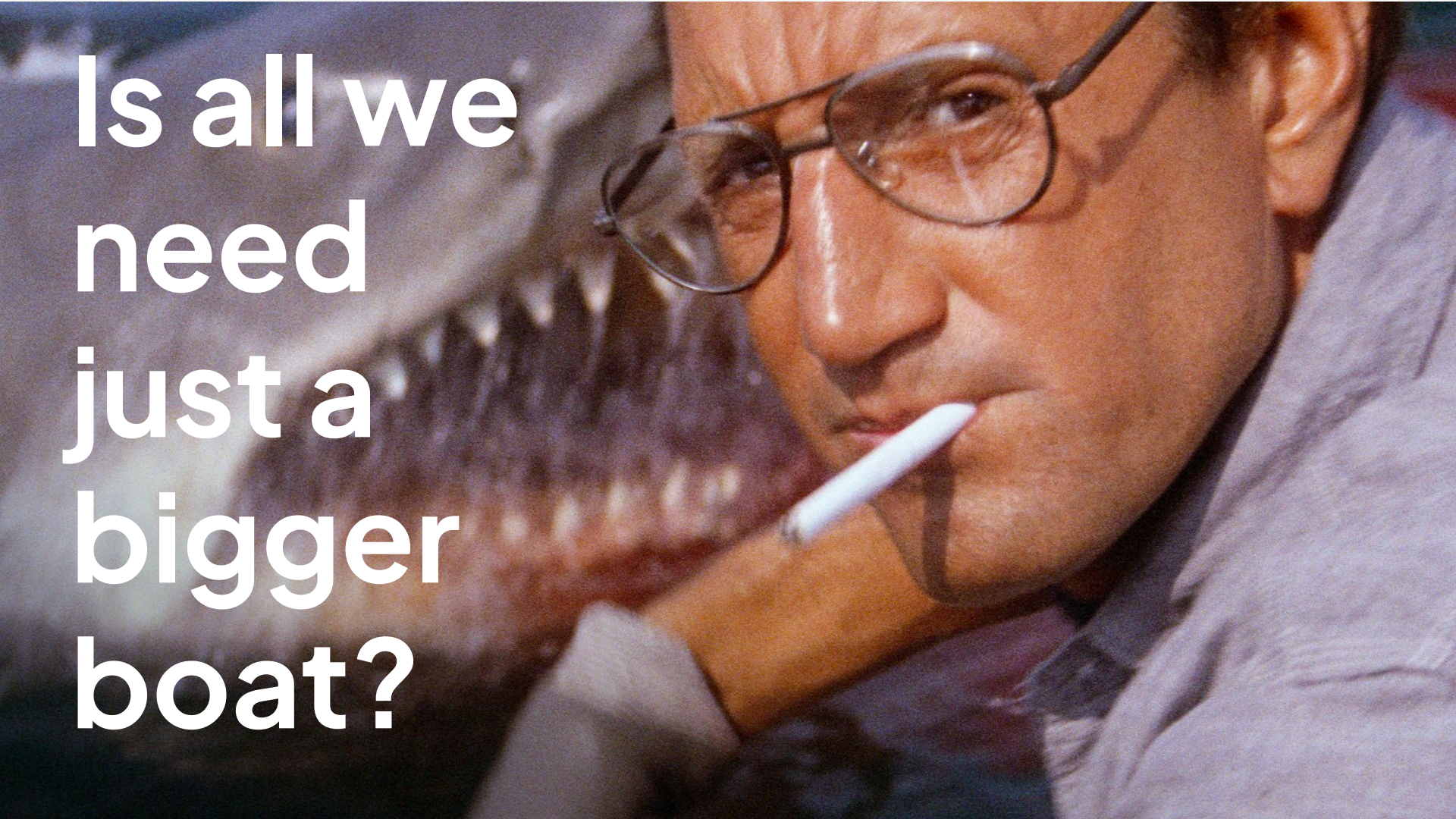
The gains are converging

March 2026 Arena Elo: Anthropic 1503 · xAI 1495 · Google 1494 · OpenAI 1481 · Alibaba 1449 · DeepSeek 1424. Four companies within 25 points.





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Is all we need just a bigger boat?

LLMs are Multitask

Task-Specific to General-Purpose

Moving away from narrow, single-task machine learning models to highly versatile foundation systems.

Multi-Domain Capabilities

Driving widespread value and automation across key domains including coding, support, content, and analytics.

Image from Huyen 2025



Figure 1-4. The range of tasks in the Super-NaturalInstructions benchmark (Wang et al., 2022).

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From models to applications

Adaptation

Reusable Foundation

Reusable “foundation” you adapt, not retrain.

Accessibility

API-Driven

API access lowers the barrier → faster product loops.

Deployment

Convenience vs. Control

Self-hosting and open models trade convenience for control.

Image from Huyen 2025

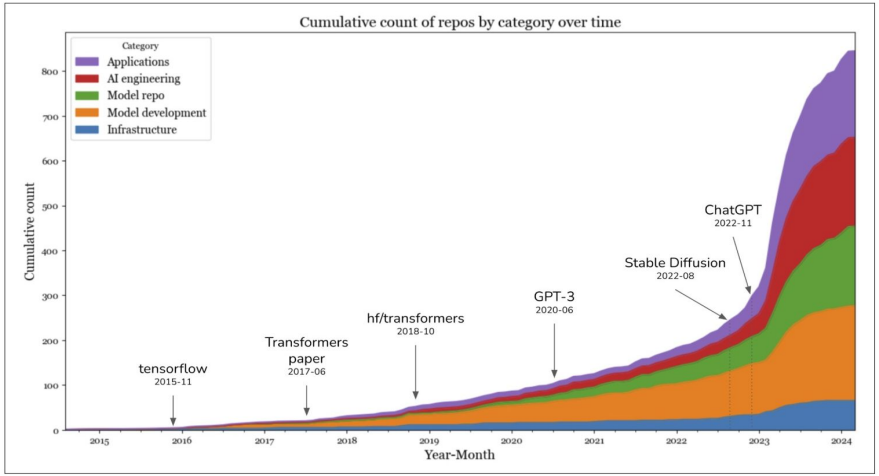


Figure 1–15. Cumulative count of repositories by category over time.

Traditional Language Models	Large Language Models (LLMs)
Trained on small datasets with limited compute → **narrow capabilities.**	Trained on massive datasets with huge compute → **qualitatively new capabilities.**
Task-specific (e.g., translation, classification, spam detection, etc.)	**General-purpose** (translation, reasoning, summarization, coding, multi-domain).
Scale to gain incremental improvements.	Scale to gain emergent abilities (reasoning, tool use, code generation).
Each model trained for a single task.	Foundation models: one model adapted to many tasks.

Take a look at:

Wei et al., Emergent Abilities of Large Language Models (2022) arXiv

Kaplan et al., Scaling Laws for Neural Language Models (2020) arXiv

Multi-modal models

- Foundation models encapsulate knowledge about the world
- And data comes in several modalities
- Historically we don’t do a lot with unstructured data
- Foundation models will increasingly be multi-modal

An example: from CLIP to Fashion CLIP

Download model:

https://huggingface.co/patrickjohncyh/fashion-clip

	Image	Textual Data	
Farfetch		Short Description	Cropped knit cardigan
		Gender	Women
		Brand	M Missoni
		Highlights	light brown/multicolour; mix print; ribbed-knit edge; V-neck; three-quarter length sleeves; ribbed hem; cropped
KAGL		Product Name	Nike Women As Element Ja Pink
		Category	Topwear
		Sub-category	T-Shirts
F-MNIST		Category	T-shirt / Top
DEEP		Product Description	Breton Stripe Henley
		Category	Henley
		Attributes	Long sleeve; Sleeve



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Contrastive language and vision learning of general fashion concepts

Patrick John Chia ✉, Giuseppe Attanasio, Federico Bianchi, Silvia Terragni, Ana Rita Magalhães, Diogo Goncalves, Ciro Greco & Jacopo Tagliabue

Scientific Reports **12**, Article number: 18958 (2022) | Cite this article

21k Accesses | **32** Citations | **14** Altmetric | Metrics

A Publisher Correction to this article was published on 23 January 2023

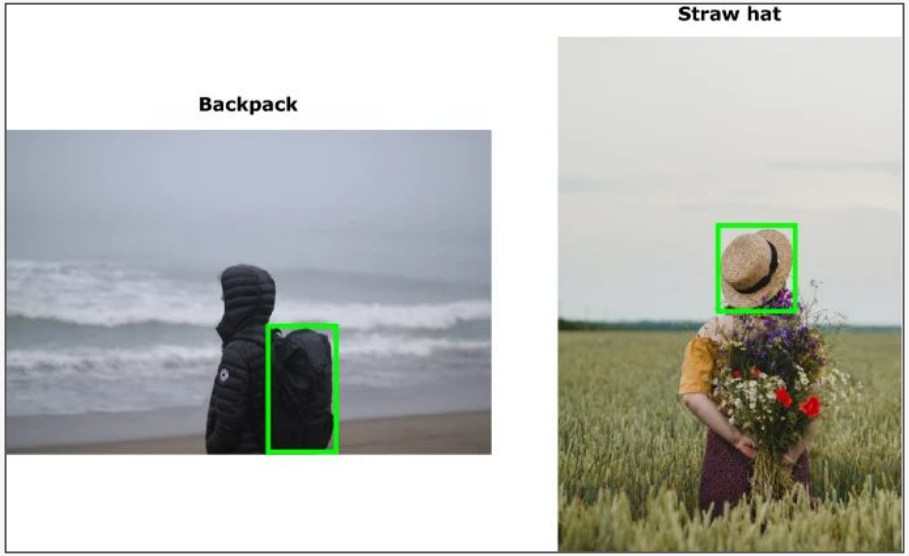
This article has been updated

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An example: from CLIP to Fashion CLIP

Download model:

https://huggingface.co/patrickjohncyh/fashion-clip





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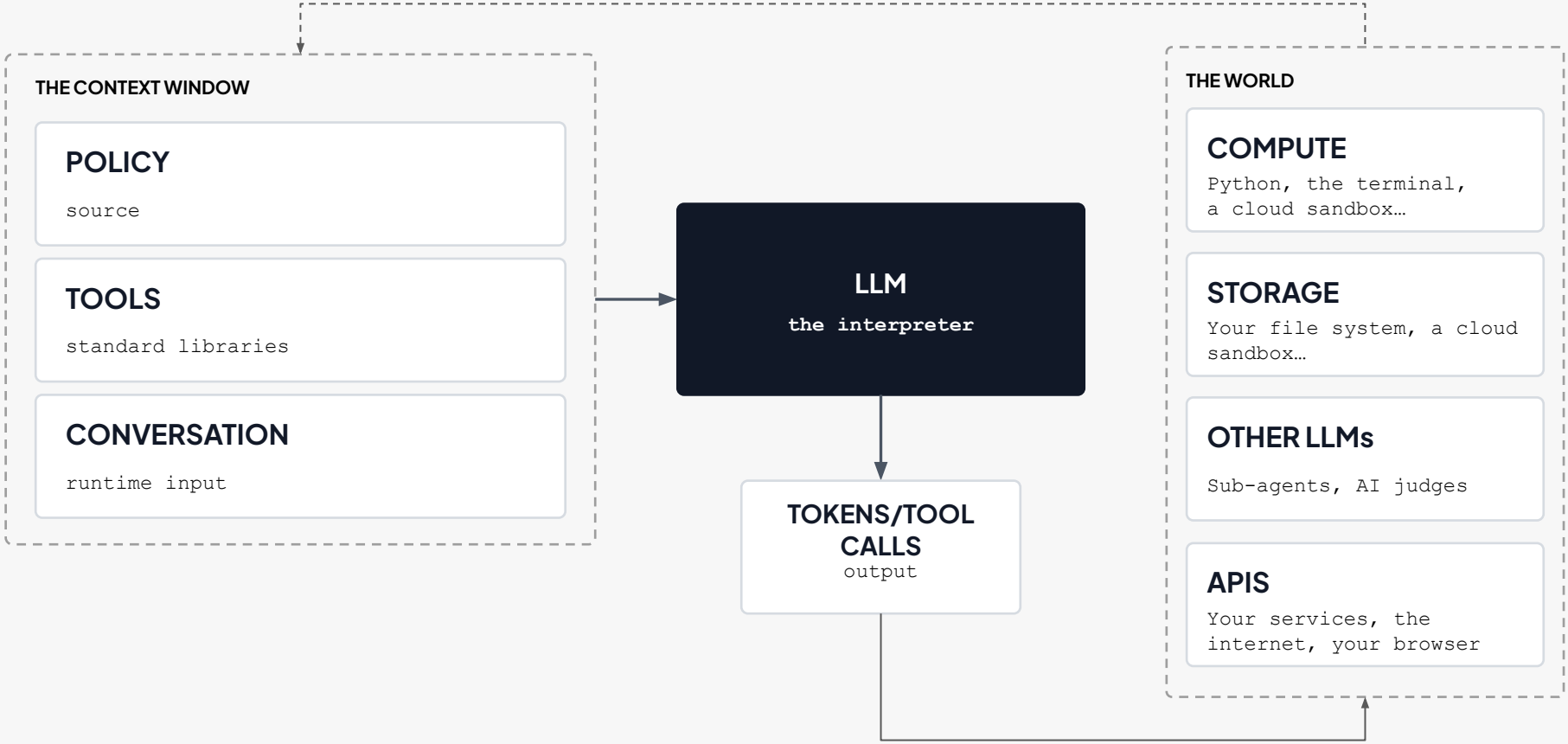
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A new form of programming

An LLM is a new kind of programmable machine



Karpathy's software 1.0 → 2.0 → 3.0

Software 1.0

```python
def simple_sentiment(review: str) -> str:
    """Return 'positive' or 'negative' based on a tiny keyword lexicon."""
    positive = {
        "good", "great", "excellent", "amazing", "wonderful", "fantastic",
        "awesome", "loved", "love", "like", "enjoyed", "superb", "delightful"
    }
    negative = {
        "bad", "terrible", "awful", "poor", "boring", "hate", "hated",
        "dislike", "worst", "dull", "disappointing", "mediocre"
    }

    score = 0
    for word in review.lower().split():
        w = word.strip(".,!?;:")          # crude token clean-up
        if w in positive:
            score += 1
        elif w in negative:
            score -= 1

    return "positive" if score >= 0 else "negative"
```

Software 2.0

10,000 positive examples

10,000 negative examples

encoding (e.g. bag of words)

↓ train binary classifier

parameters

Software 3.0

You are a sentiment classifier. For every review that appears between the tags
<REVIEW> … </REVIEW>, respond with **exactly one word**, either POSITIVE or NEGATIVE (all-caps, no punctuation, no extra text).

Example 1
<REVIEW>I absolutely loved this film—the characters were engaging and the ending was perfect.</REVIEW>
POSITIVE

Example 2
<REVIEW>The plot was incoherent and the acting felt forced; I regret watching it.</REVIEW>
NEGATIVE

Example 3
<REVIEW>An energetic soundtrack and solid visuals almost save it, but the story drags and the jokes fall flat.</REVIEW>
NEGATIVE

Now classify the next review.

From Karpathy's YC lecture

Karpathy's software 1.0 → 2.0 → 3.0

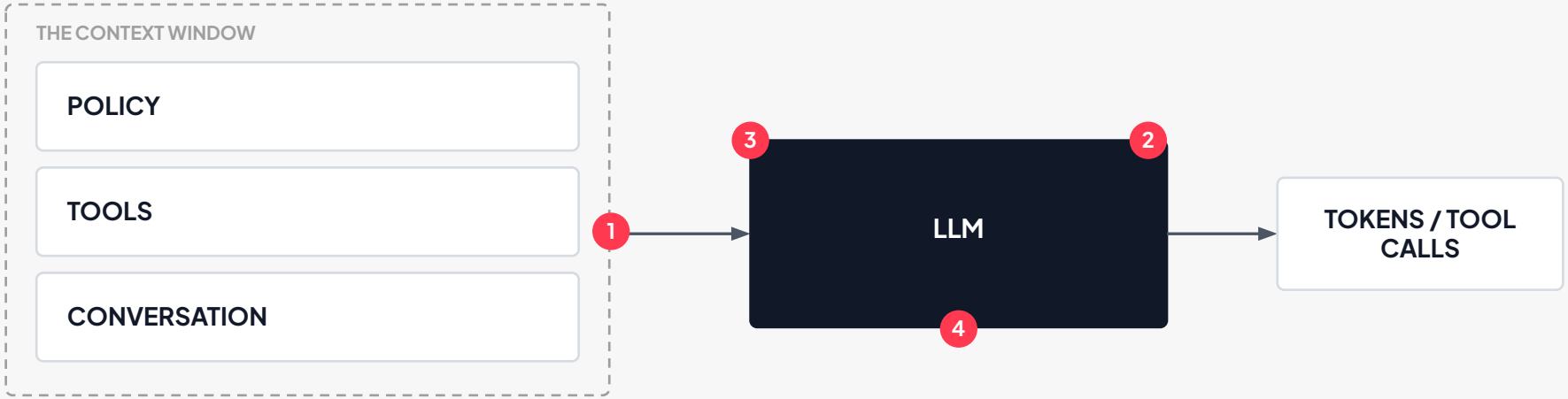
	Software 1.0	Software 2.0	Software 3.0
Users write	Instructions	A dataset and an objective	Context, in English
The artifact is	Source code	Weights	A prompt
Executed by	A CPU deterministic	A forward pass fixed once trained	A model probabilistic
Rule	`if (today - delivered) .days > 30: return REJECT`	40,000 labelled past decisions → a classifier	*"Returns are accepted within 30 days of delivery."*

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What you gain when you give things up

	Software 1.0	Software 2.0	Software 3.0
You gained	Automation	Problems you cannot specify	No training run. Change behaviour in seconds.
You gave up	—	The ability to read easily the program	The ability to predict deterministically the program
What checks it?	Compiler · type system unit tests · code review	A held-out test set	**An eval set.** Nobody gives you one. You write it. → Week 3

The abstractions and the harness are still pretty coarse



1 **No type system.** Nothing checks the program before it runs. There is no syntax error — only a wrong answer that reads well.

2 **1,000 runs at temperature 0 → 80 different answers.** Example, kernels reduce differently at different batch sizes so your output depends on who else is calling the API even if temperature is 0.

3 **The interpreter changes under a stable name.** Example: in 2023, GPT-4 got weirdly worse at identifying primes: 84% → 51% in three months. Same identifier, no notice.

4 **Pinned snapshots still expire.** gpt-4-0613 shutted down 23 Oct 2026.

AI engineering

The AI engineer

Three layers of responsibility

01

Application Development

Prompt design, context management, user interfaces, and evaluation loops.

02

Model Development

Model choice, post-training, dataset engineering, and inference optimization.

03

Infrastructure & LLMOps

Serving models, managing compute, monitoring, scaling, and overall performance.

AI Engineering

How core responsibilities map to concrete techniques and workflows

Work with foundation models trained elsewhere.	→	**Application Layer** **Prompt & context engineering, sampling, fine-tuning**
Handle compute-intensive models, GPU scaling, efficiency, inference engineering.	→	**Operations Layer** **LLMOps**
Manage open-ended outputs → much harder evaluation, testing and guardrails.	→	**Quality & Trust Layer** **Eval, Eval, Eval!**

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Model adaptation and application development

Closer to product & full-stack work

01

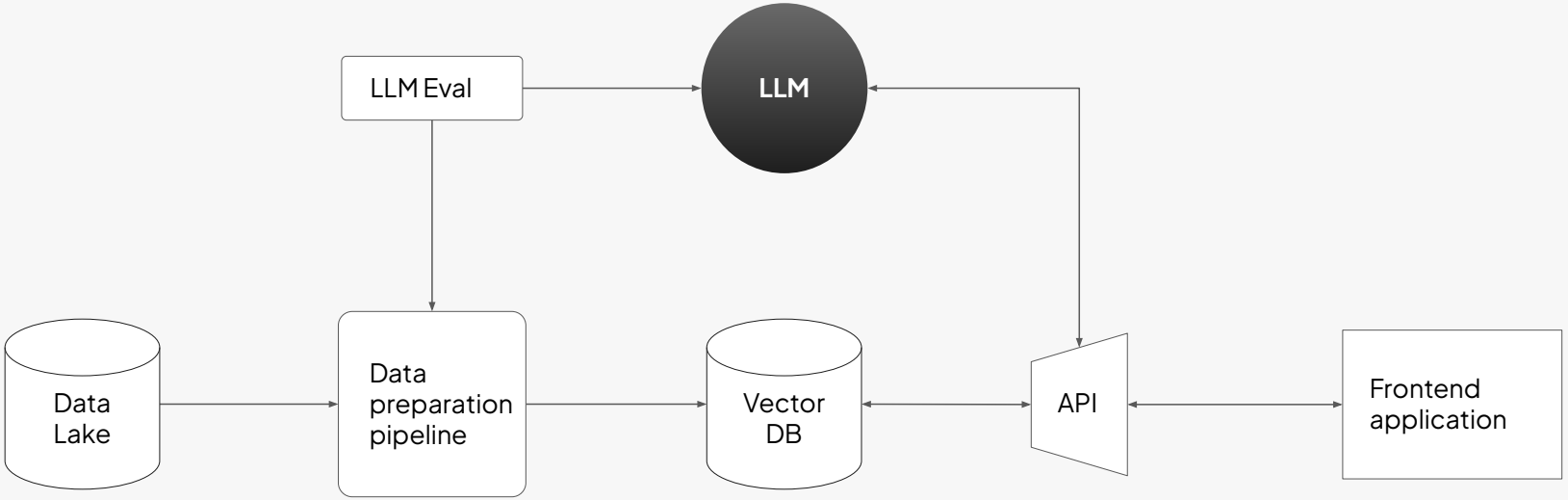
Product Mentality

AI engineers increasingly collaborate on UX, feedback loops, and product decisions.

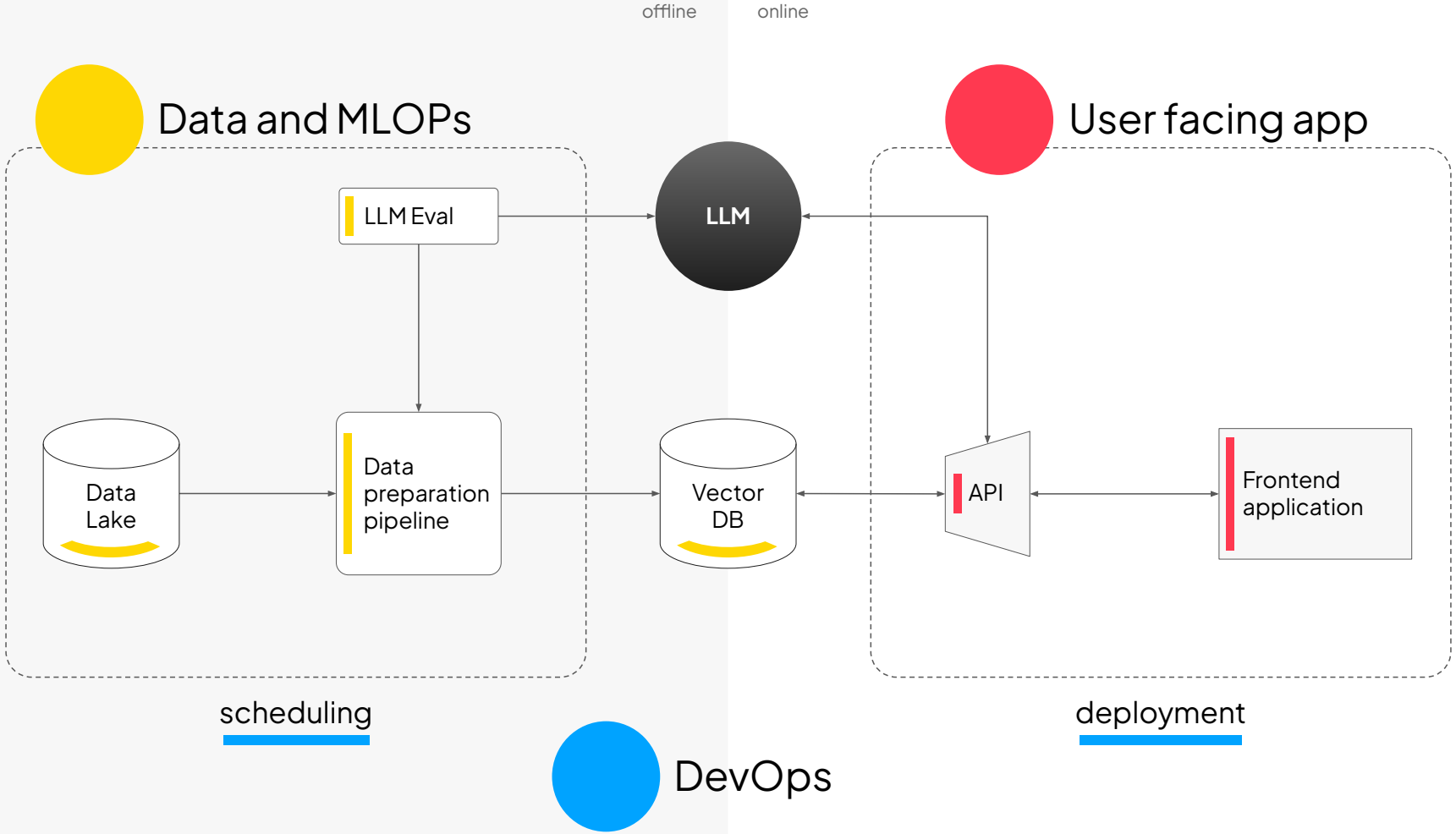
02

Full-Stack Backgrounds

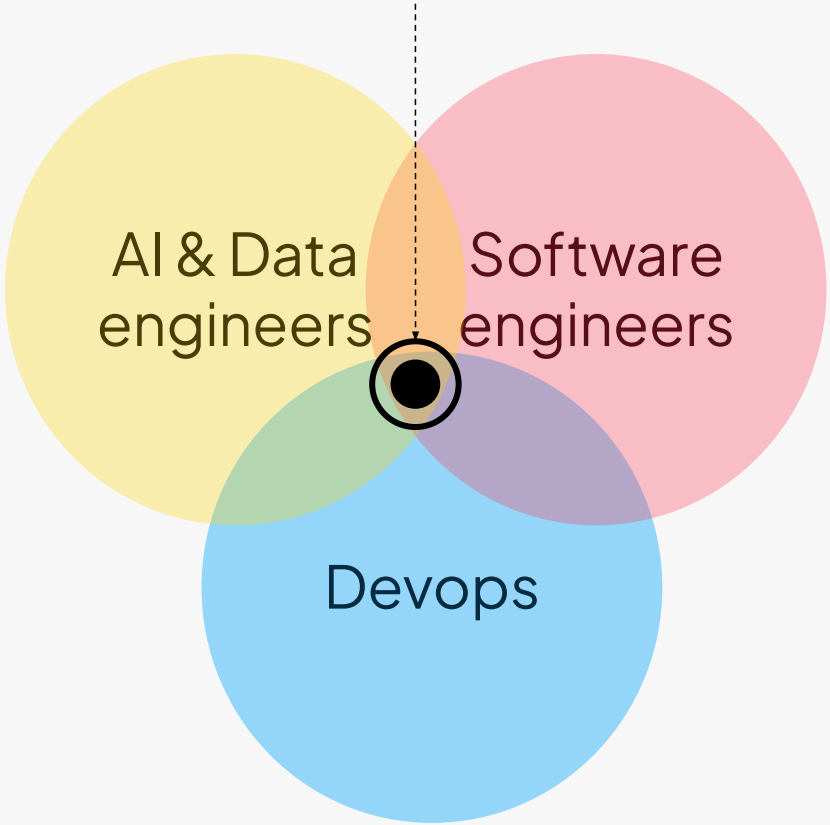
Many come from web dev / full-stack backgrounds, not only ML.



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offline
online
Data and MLOPs
User facing app
LLM Eval
LLM
Data Lake
Data preparation pipeline
Vector DB
API
Frontend application
scheduling
deployment
DevOps

Your AI application!



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Historically, productionizing ML has been pretty hard

I am a data scientist, I work with csv files and notebooks



Data

Notebooks on my laptop and Python scripts

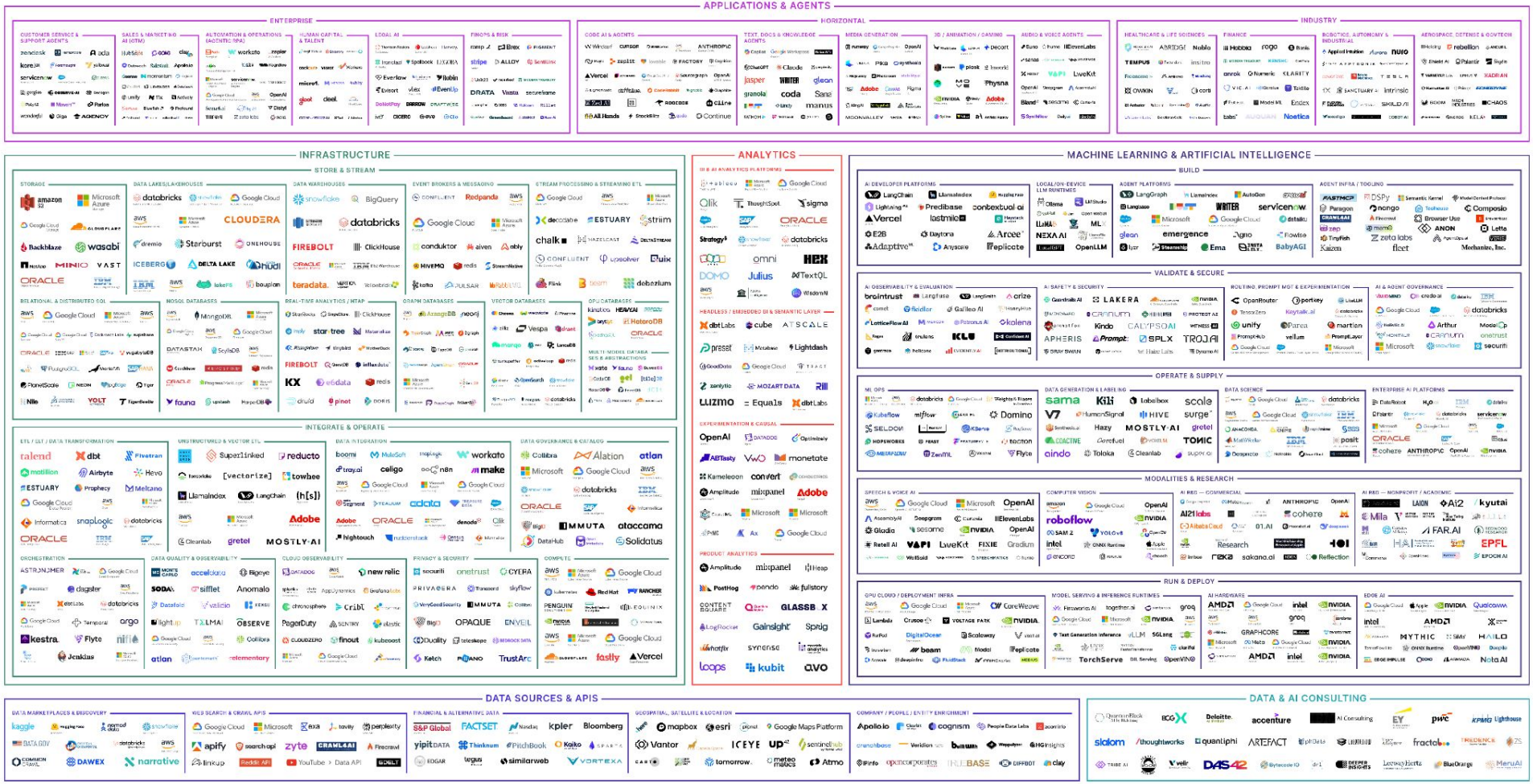


Software and DevOps Engineers

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THE 2025 MAD (MACHINE LEARNING, ARTIFICIAL INTELLIGENCE & DATA) LANDSCAPE

MAD.FIRSTMARK.COM



Paradigm shift in product development

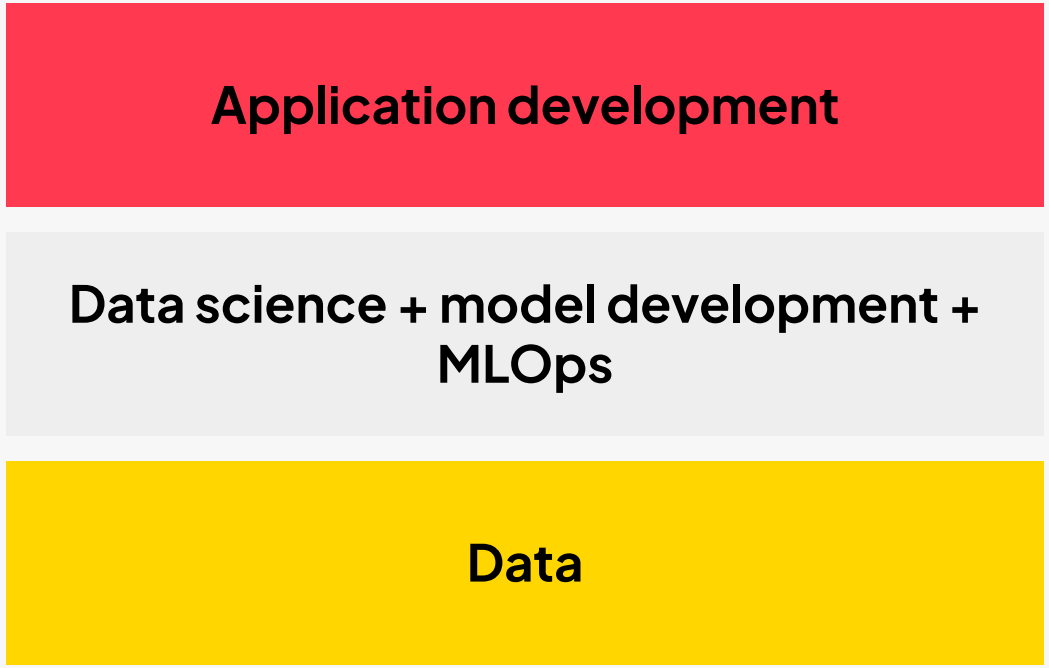
Circa 2015–2023*

01 Invest in data
02 Invest in Analytics
03 Invest in Data Science
04 Invest in ML and AI
05 Deploy AI applications
06 Measure ROI

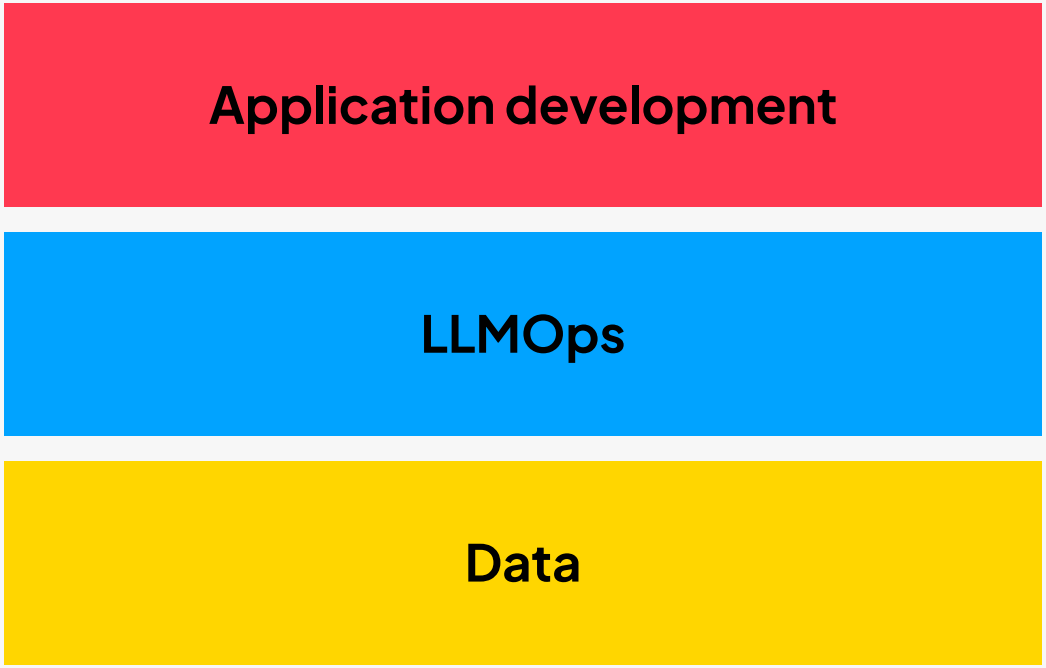
Today

01 Put together a prototype
02 Measure rough ROI
03 Invest in data
04 Invest in Analytics
05 Invest in Data Science
06 Invest in ML and AI

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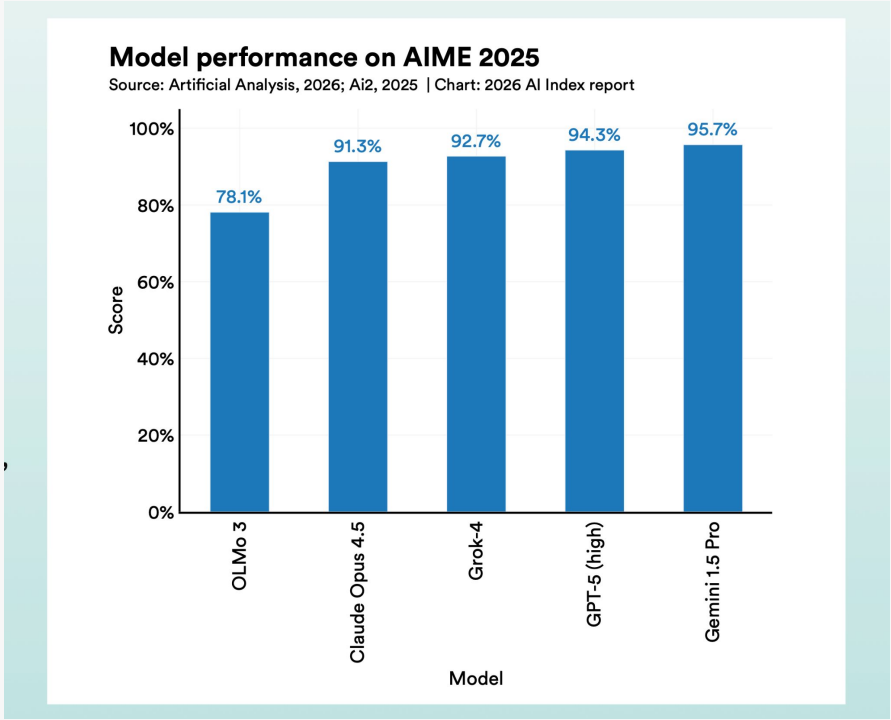
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Data centric AI

- **Data quality** - if AI has high ROI for you and you have some special that only you have, using OpenAI won't cut it.
- **Data infrastructure** - data infrastructure is historically super hard and an IT domain. Simplify and put it in the hands of application builders.
- **Data strategy** - data touch engineering, product, security, compliance and legal. You might want to think about it thoroughly.



Why should you care about data?

01

Differentiate

Use data specific to your business.

02

Integrate

Use accurate, up-to-date information.

03

Govern

Ensure auditability, compliance, and trust.

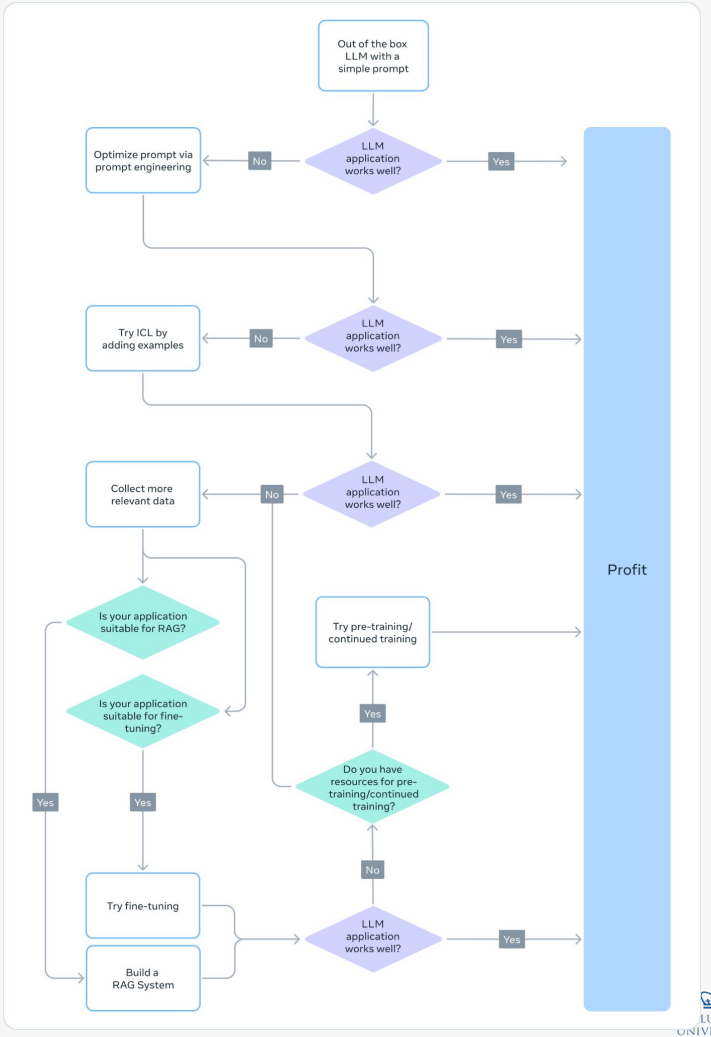
04

Improve

Use feedback data to improve AI systems.

Buy vs. build

- Thin slicing: start simple but always build end-to-end to quickly determine ROI.
- Spend as much time as you can understanding the use case to identify weak vs. strong constraints.
- Do not - I repeat **do not** - do what Google does. Assume you'll operate at a reasonable scale.



Source: https://ai.meta.com/blog/adapting-large-language-models-llms/

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Practical information

Course logistics

- By week 2 - form groups of min 3 max 5 people. Self-organize or you will be organized.
- By week 3 - present your idea for a Capstone project and send it to me.
- By week 5 - send the repo with your first draft of the Capstone project.
- By week 7 - push the final version of your Capstone project to your repo.